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Stabilization of a quadcopter drone by a PID loop.

The quadcopter drone spectacularly illustrates the mechanical instability of a physical system. Its dynamics show oscillations that naturally tend to diverge, due to gravity and inertia. The PID loop offers a way to study how appropriate control can reduce these oscillations and stabilize the system.

The quadcopter drone highlights the theme "Cycles, loops": its dynamics naturally produce divergent oscillatory cycles (mechanical instability). The PID loop (continuous feedback) damps these cycles and stabilizes the system, illustrating how a loop controls and suppresses unwanted cycles.

The candidate attests to having worked alone.

Thematic positioning (STEP 1):

- *MATHEMATICS (Analysis)*
- *PHYSICS (Mechanics)*
- *PHYSICS (Interdisciplinary Physics)*

Keywords (STEP 1):

Keywords (in French)	**Keywords (in English)**
Quadricoptère	*Quadcopter*
Contrôleur PID	*PID controller*
Boucle de rétroaction	*Feedback loop*
Cycles oscillatoires	*Oscillatory cycles*
Dynamique du roulis	*Roll dynamics*

Annotated bibliography

Quadcopter drones were developed for a variety of applications, ranging from surveillance to aerial photography, including search and rescue. Indeed, their many advantages, such as maneuverability, low power consumption and controlled stability, allow their use in several fields. However, in recent years, PID control systems have enjoyed renewed momentum: they are used as regulators to stabilize mechanically unstable systems, such as quadcopters, by measuring angular deviations caused by thrust variations or external disturbances. This is exploited in aeronautics, robotics and transportation [1], [2].

Indeed, the study of the information carried by differential equations, stability and damped oscillations makes it possible to retrieve information about the physical dynamics: mechanical disturbances modify the angular paths followed by the axes of rotation. These modifications can then be detected by observing the Laplace transform. Moreover, feedbacks arise from reflections due to inhomogeneities in the PID gains, which also make it possible to trace back variations in stability or oscillations [3], [4].

The PID, or Proportional-Integral-Derivative controller, is an automated regulation mechanism used to

maintain a physical quantity at a setpoint value. It computes a correction based on the current error (P), the accumulation of past errors (I) and the anticipated future variation (D). This eliminates persistent deviations and damps oscillations, turning an unstable system into a stable one [1], [3].

The operation of the PID relies on a continuous feedback loop: measuring the error, computing the three terms and applying the command to the actuators. The P term reacts proportionally to the error, I integrates to correct constant biases, and D differentiates to anticipate and damp. In a drone, this stabilizes the angles (roll, pitch) against disturbances such as wind [2], [5].

It is possible to build a purely analog PID from laboratory components, such as operational amplifiers, resistors and capacitors. For example, an RC integrator for the I term, a CR differentiator for D, and a summing amplifier to combine the signals. These simple circuits, with op-amps such as the TL081, reproduce the PID loop and make it possible to test anti-windup via diodes [3], [6].

The mechanical study of the drone reveals nonlinear dynamics linearized around hovering flight, with Newton–Euler equations for roll: $I_x \ddot{\varphi} = \tau_\varphi$. Without PID, the system is unstable with divergent oscillations. The PID introduces a third-order equation, shifting the poles to damp the cycles and ensure stability [4], [7], [8].

To conclude, the PID loop is a means of analyzing and stabilizing the dynamic environment of a drone, and many studies are under way to improve its robustness against disturbances.

Selected research question

§How can a PID loop capable of stabilizing a quadcopter drone be designed with simple laboratory components, and to what extent does tuning this loop make it possible to control the oscillatory cycles of its dynamics?

Candidate's TIPE objectives

General objective:

Show how a PID loop stabilizes an unstable quadcopter drone by controlling its oscillatory cycles, within the theme "Cycles, loops".

Specific objectives:

Explain the role of the P, I, D terms.

Build an analog PID with op-amps and lab components.

Model the roll dynamics $I_x \times \ddot{\varphi} = \tau_\varphi$.

Analyze the effect of the gains on the damping of oscillations and on stability.

Relate theory and experiment to turn a divergent system into controlled flight.

Bibliographic references (STEP 1)

[**1**] GENE F. FRANKLIN AND ABBAS EMAMI-NAEINI: Feedback Control of Dynamic Systems *: Pearson, 2019*

[**2**] KATSUHIKO OGATA: Modern Control Engineering *: Pearson, 2010*

[**3**] NORMAN S. NISE: Control Systems Engineering *: Wiley, 2019*

[**4**] TIMOTHY W. MCLAIN, RANDAL BEARD: Small Unmanned Aircraft: Theory and Practice *: Princeton University Press, 2012*

[**5**] LIUPING WANG: PID Control System Design and Automatic Tuning using MATLAB/Simulink *: Wiley, 2020*

[**6**] SAMIR BOUABDALLAH, ROLAND SIEGWART: Backstepping and Sliding-mode Techniques Applied to an Indoor Micro Quadrotor *: IEEE ICRA, 2005*

[**7**] ROBERT MAHONY, VIJAY KUMAR, PETER CORKE: Multirotor Aerial Vehicles: Modeling, Estimation, and Control of Quadrotor *: IEEE Robotics & Automation Magazine, 2012*

[**8**] LESZEK CEDRO, KRZYSZTOF WIECZORKOWSKI: Optimizing PID controller gains to model the performance of a quadcopter *: Transportation Research Procedia, 2019*